Manual for Geuldal_NBS_simulations

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Introduction

The code in this repository is developed to setup a database and simulate the impact of Nature Based Solutions (NBS) on floods in the Geul river catchment. The model simulations are done with the OpenLISEM model.

The goal of this code is to speed up the preparation of model runs, make a reproducible workflow and give a flexible platform to extend and further use the OpenLISEM model for the Geul catchment.

The aim of this document is to explain the different options of using the code. The first chapter guides through the preparation to install software and download data. In Chapter 2 the different routes to make actual OpenLISEM runs are explained. Chapter 3 shows how to add additional NBS or scenario's to the workflow and in the last chapter some indications for advanced usage are given.

WARNING many sections in the code are automated to speed up and simplify the process of making the OpenLISEM runs. However, if wrong input is given, this can result in errors – please make sure you follow the exact steps. Besides that, working with code and models often includes trying to understand why things don't go as expected. This repository is given without a guarantee of always producing results or working without errors. The first time installation and setup of all libraries in R and RStudio can be challenging. See comments in section 2.2 for some more explanation. If you get stuck, try to solve the problem, or send an email to one of the authors/contributors.

1. Setup and software installation

For the code to work you need the following software / data:

1. The Geuldal_NBS_simulations code
2. The 'spatial_data' folder.
3. The OpenLISEM model
4. conda with the packages pcraster and python installed
5. R and Rstudio

In addition when you aim to do more in depth modelling QGIS and Nutshell might be usefull programs to also install.

1. The Geuldal_NBS_simulations code.

The code can be obtained by downloading the code from https://github.com/mcommelin/Geuldal_NBS_simulations

If you know Git you can clone the repository, or you can download the code and spatial data from the releases page, often the latest release is the best choice.

https://github.com/mcommelin/Geuldal_NBS_simulations/releases

Download the source code and extract the folder if needed.

2. The spatial_data

The latest version of the spatial_data can also be downloaded from the release page:

https://github.com/mcommelin/Geuldal_NBS_simulations/releases

Extract the spatial data and place it inside the folder with the downloaded code.

The path should be ./Geuldal_NBS_simulations/spatial_data/

3. The OpenLISEM model

For Windows devices:

1. Go to: https://github.com/vjetten/openlisem/releases/
2. Download 'openLlisemSetup_7.4.9.exe'; this will download an installer for the OpenLISEM model
3. When downloaded, double click the downloaded file on your device and follow the instructions. A warning about installing unknown software might pop up. Click 'more info' and then 'Run anyway' to install OpenLISEM.
4. When finished the LISEM model can be found in your software section. Besides the model also the Nutshell software is installed.

For users of Linux or IOS, try using a virtual machine with windows. For Linux compilation intructions are available in the OpenLISEM repository.

4. conda with packages pcraster and python

To setup PCraster and python in a good way we need some extra steps. First we will install the program 'miniconda'; in this program we can then install additional packages with a command.

NOTE: if you already have (a version of) conda installed on your device, you can skip installing miniconda. Double installations sometimes also give errors.

1. Download miniconda from: https://docs.conda.io/en/latest/miniconda.html
2. Select the Windows 64-bit
3. Run the downloaded .exe file and follow the instructions.
4. In the tab 'Choose Install Location' select a folder location where you have write acces. For example C:/Users/[username]/
5. All other settings can be left to the default

After the installation open 'Anaconda Prompt (Miniconda3)' from the start menu. A command window will open. With a line staring with:

```
(base) C:\Users\[username]>
```

In conda we create a new environment with two packages we need.

a. Type the following command:

   ```
   conda create --name lisem -c conda-forge
   python=3.13 pcraster
   ```

   This command will create a new environment with the name 'lisem' and install the packages pcraster and python in the environment.

b. Miniconda will suggest a long list of packages to install and ask:

   ```
   Proceed ([y]/n)?
   ```

   Answer with: `y` and hit 'enter'
   This will install all packages that are needed to use PCRaster

If you want more information, this can be found on: https://pcraster.geo.uu.nl/pcraster/4.4.2/documentation/pcraster_project/install.html

5. R and RStudio

R is a scripting language that we use in this project to handle data, do spatial calculations and prepare the input databases for OpenLISEM. RStudio is a program that gives a more convenient user interface to work with R.

R can be downloaded from: https://cran.r-project.org/bin/windows/base/

And RStudio from: https://docs.posit.co/ide/user/#rstudio-ide-oss-downloads

Install both programs when downloaded.

Now you should be ready to start making the first simulation databases for OpenLISEM.

2 Prepare OpenLISEM simulations

Within the code there are three options to work through all steps to make OpenLISEM simulations:

1. Interactive workflow: Based on questions you are guided trough all steps and the desired OpenLISEM runs are created for you. Works well for new users, and if you don't want to make to many runs.

2. Manual step-by-step: This is the route which was used during the developement of the whole model setup. All parts of the code can be adapted and executed by hand. Most usefull for advanced users and for testing of new features.

3. HPC workflow: an automated – script based workflow to prepare OpenLISEM simulations for the whole Geuldal which can be executed on a HPC or in batch mode.

2.1 The interactive workflow

If you are new to OpenLISEM simulations for the Geuldal, this is the best place to start. Make sure you have donwloaded and installed everything as described in chapter 1.

Figure 1 shows a folder layout as it should look also at your device:

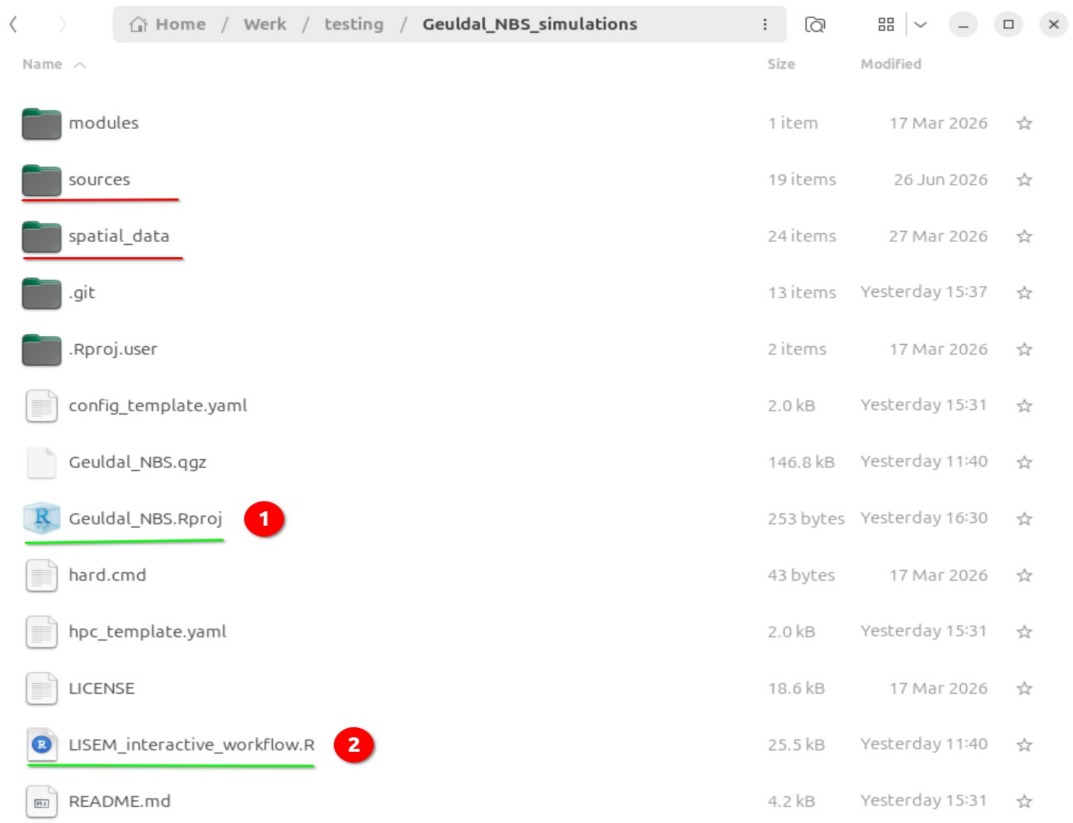

Figure 1: Repository after download, the spatial_data folder is placed inside.

The spatial_data folder should be placed inside the repository with the code. Inside the repository there is also a folder called 'sources', here you will find all scripts, but also tables with choices and settings which are used to setup the OpenLISEM databases.

1. To start the interactive workflow, find the RStudio project file 'Geuldal_NBS.Rproj' and double click this. An Rstudio project file helps to link scripts within a code repository to each other.

2. Next also double click the file 'LISEM_interactive_workflow.R'. This will open a file in the upper left window of RStudio. Your Rstudio will look something like this:

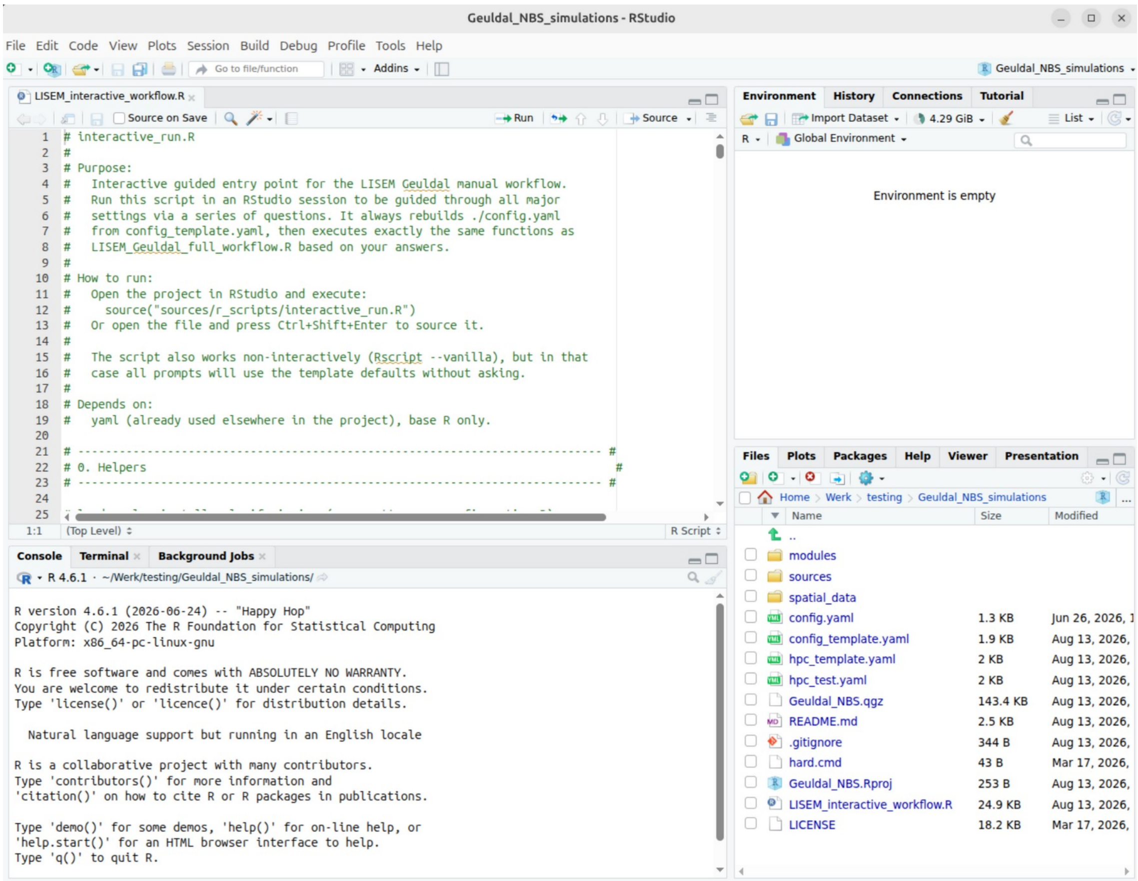

Figure 2: RStudio user interface, with in the top right panel the interactive workflow script opened.

NOTE: A warning about missing packages might pop up, but this will be handeld within the code, so can be ignored for now.

3. To make the code work smoothly, we need to change some settings in RStudio. Go to Tools→Global options…

A screen will open. Under the tab General, uncheck the box Restore .Rdata into workspace and set the Save workspace option to never. Click ‘Apply’ and ‘OK’

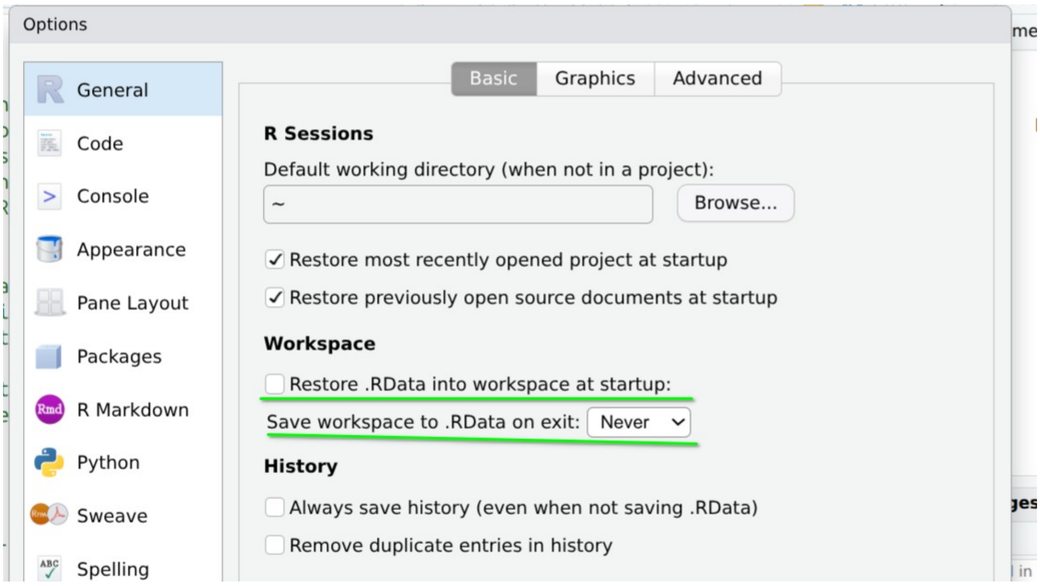

Figure 3: Global options in RStudio, with proposed adjustments in the settings

4. Next 'source' the opened script, this will execute all code in this script and it will guide you through the further setup and preparation of the data. You can find the 'source' button in the top right of the screen panel with the opend file:

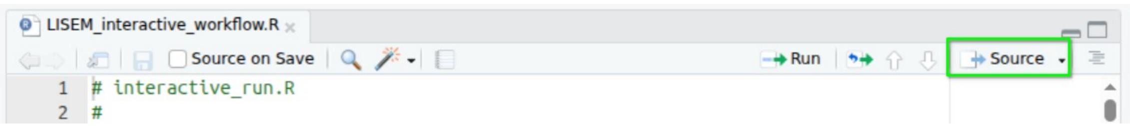

Figure 4: Source button in Rstudio, this will execute the code in the opened script.

NOTE: sometimes installing a packages automatically and directly using it gives errors. If you get an error about YAML, restart RStudio – which normally solves the error.

When pressing source, the first question of the interactive setup should pop up in the console – the lower left pane in RStudio:

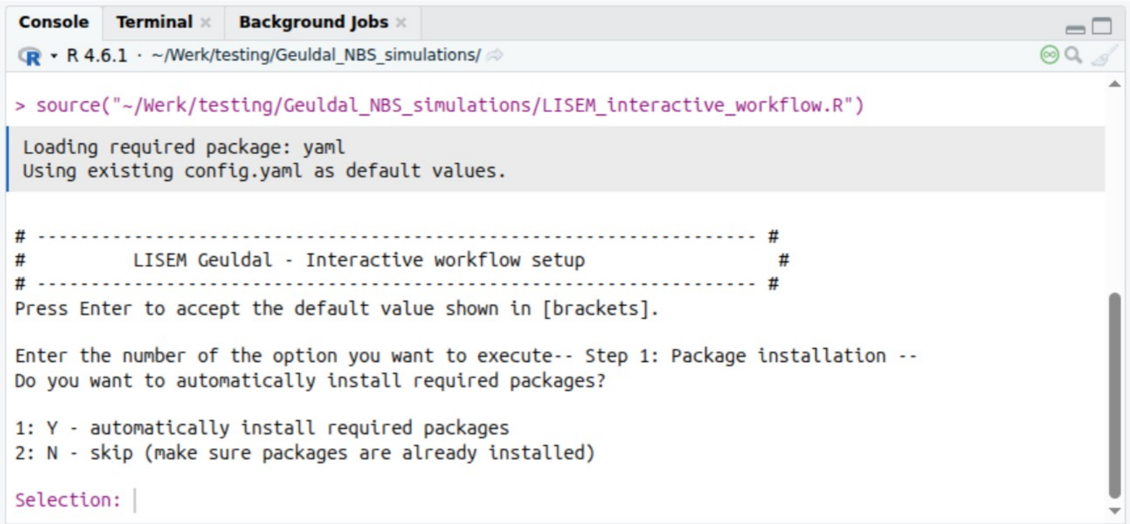

Figure 5: RStudio lower left panel, with the first question of the interactive workflow.

When you get a question with multiple options, enter the number of the option you want to execute and hit enter. Below for some questions additional explanation is given.

Q3: (mini)conda path, you have to fill the path to the location where (mini)conda is installed. This can be something like "C:/Users/[name]/miniconda3". The folder should be available on the location you installed it in chapter 1.

> **Note:** In windows a path is written with '\' as separator. R does not understand this, so replace these for '/'!

Q4: Conda environment, you have to fill the name of the conda environment you created. The option between brackets [lisem] is the default. If you also named the environment 'lisem' you can hit enter to use the default, otherwise type the name of the environment you installed pcraster and python in.

After this question the packages in R will be installed / loaded. If this is the first time you run the code, this might take some time.

> **Note:** the automatic installation of libraries on different devices is often not very smooth. The current code is tested for Windows and should work, however sometimes the process might get stuck during installation of libries and give errors.
> If this occurs. Please close RStudio and reopen the interactive workflow. Often restarting will solve the problems.

If all libraries installed correctly the next question will pop up, now the actual preparing of OpenLISEM simulations can start!

The next question (Q5) is what type of simulation you want to prepare. Data_preparation is automatically included in all options, so you don't have to run this separatly.

Note: option 4: simulate scenario, only works if you added scenarios as described in chapter 3. Otherwise choose option 2 or 3.

Q8: NBS simulations currently 11 different measures are available, the spatial implementation in this setup is based on the maps made by Stroming for WRL. Type 'list' as answer to the question to see which measures are available. If you want to add maps with single NBS, see chapter 3 for instructions.

NBS scenario's: Currently only a hypothetical test scenario is available (101). Real scenario's must be added by users. If you want to add a scenario map yourself, see chapter 3 for instructions.

Because the code is developed in close cooperation with the WRL project, there are two possible formats of land use classes. A landuse class is a number on the map that corresponds to a type of landuse which characterises many parameters in OpenLISEM. The two options are 'def' = default and 'wrl' = WRL specific format. In figure 6 the description of the landuses and the corresponding numbers are given. Only landuse classes that occur in the default list can be simulated with OpenLISEM. If other numbers are provided, they are replaced in the code by the original landuse of the base simulations for the Geulcatchment.

	lu_classes	
beschrijving	wrl	def
Structuurrijk_grasland	1	11
Afleiden_water	4	
Voedselbos	6	14
Natuurlijk_bos	7	15
Structuurrijke_bossen	8	16
Contourgreppels	9	17
Infiltratiestroken	12	20
Checkdams	15	
Beekbedding_verondiepen	17	
Dalbodem verruwen	18	
Houten_checkdams	19	
Akker	101	1
Grasland	102	3
Bos	103	2
Verhard	104	5
Water	105	6
Natuurlijk gras	106	4
Inundatiegebieden	202	
maatregel x	203	
wegen	301	
Akker_teeltmaatregelen	999	

Figure 6: available landuse class numbers for WRL and for default. ***Note!*** *this tables is not finished yet, in coorperation with WRL all NBS still need to be included.*

Q10: The number of cores you want to use for OpenLISEM depends on the device you use. Using all cores might slow down the pc, and with the user interface of OpenLISEM might give instable performance. Often 4 or 8 cores works well, in general more cores result in faster model runs.

2.2 Manual step-by-step

The manual workflow can be started by using the script '/sources/r_scripts/LISEM_Geuldal_full_workflow.R'

There are many comments inside this script explaining choices and setup. Besides that, by following the 'source()' functions, the locations of additional scripts with specific actions (e.g. make swatre infiltration tables, or make the OpenLISEM runfiles) can be found.

The infitial settings used for configuration have to be stored in a config file (default = config.yaml). Open the file config_template.yaml in a text editor or RStudio and set all setting to the values you want, save as config.yaml to make a smooth start.

2.3 HPC workflow

To be able to make many simulations, and to simulate the whole Geulcatchment. An automated workflow can produce runs that are suitable to simulate with a HPC and/or in batch mode.

In several ways this workflow differs from the first two approaches.

1. For the simulations of the whole Geul catchment, the main rivers are excluded from the simulations. OpenLISEM is mainly suitable for hillslope processes, and this setup is designed to couple the OpenLISEM results to a river model. For example D-Hydro.

2. To speed up simulations, the Geulcatchment is devided in ~140 subcatchments, these sometimes correspond to the subcatchments as used in the manual workflow, but often differ. This is needed to cover the whole Geul catchment, but also because main rivers are removed from the modelling domain.

3. The subcatchments all get a number, which is not identical to the manual workflow. Each subcatchment number is linked to an inflow point of the river system, which can be used by a river model. The table linking subcatchment numbers to river points can be found at:
   ./sources/setup/hpc/subcatch_id_links.csv
   Besides that, a map layer is available to see the subcatchments:
   ./spatial_data/hpc_maps.gpkg layer=lisem_subcatch. The map layer contains an attribute table which is equal to the csv file
   subcatch_id_links.csv

Using the HPC workflow

1. To use the hpc_workflow, R and a conda environment with python (version 3.13) and pcraster are required.

2. Place the 'spatial_data' folder on the <u>same</u> level as the code repository instead of <u>inside</u> for the manual workflow. This is a design choice to make containerized workflows possible.

3. Adjust the settings in hpc_template.yaml and save under a different name.

4. Run the choices made in the yaml file with:

```
Rscript –vanilla ./sources/r_scripts/hpc_workflow.R
[path_to_yaml].yaml
```

This will start installing all libraries, setup all data and prepare all the OpenLISEM simulations as selected in the yaml file.

Note: in the HPC workflow only 1 measure or scenario can be prepared per call of the hpc_workflow.R script. Run multiple times to prepare more datasets.

3 Adding new NBS or scenario's to the workflow

3.1 Adding a single NBS measure

Within the current workflow the following NBS measures can be added to an OpenLISEM simulation:

Table 1: Available NBS in workflow, including number and if the NBS uses additional landscaping code.

Number	Description	do_LE
11	Extensive grased graslands	F
12	Species rich hayland	F
13	Production graslands	F
14	Food forest	F
15	Natural forest	F
16	Changing plantatins to natural forest	F
17	Swales	T
18	Vegetation strips in flowpaths	F
19	Terraces / lynchets	T
20	Infiltrationstrips	T
21	Farmponds	T

However the spatial setup of these measures can differ. The NBS numbers 11 to 21 implement the measure acording to maps provided by Stroming for the WRL project. Below we describe the different options for adding a NBS measure to the OpenLISEM simulations.

1. Adding a different spatial schematisation of 1 of the existing measures
2. Adding a new NBS which only affects landuse parameters
3. Adding a new NBS which also affects the DEM or other aspects of the landscape.

Note: This project is shared in the public domain for others to use. If you add a new NBS, consider creating a pull request with GIT, or contact the maintainer of the repository. So your contribution can also again be used by others!

3.1.1 Adding a different spatial schematisation

The maps by Stroming are located in the folder './spatial_data/NBS_maps/'. In the workflow the maps are identified based on the starting number, which corresponds to the NBS number in the table that provides more information on modelled characteristics. This table is located in 'sources/setup/tables/lu_NBS_tbl.csv'

To update the spatial schematization follow these steps:

1) Prepare a TIF map with a schematisation of the measure.
    a) The extent of the map should be equal to the maps provided by Stroming. An option to make good maps is by masking them with the mask_5m.map available in ./spatial_data/
    b) Values in the map should be as follows:
        1) If do_LE in table 1 is FALSE: 0 = no NBS, 1 = NBS
        2) If do_LE is TRUE: 0 = no NBS, 1 = fields where NBS can be applied, 2 = actual specific location of the NBS.
2) Place the map in the folder ./spatial_data/NBS_maps/
    a) make sure the map starts with the correct number corresponding to table 1. E.g. a map with foodforests should start with '16_xxxx.TIF'

b) in the folder ./spatial_data/NBS_maps only one map with each number can be placed. Otherwise the workflow will give errors.

3) If do_LE is TRUE, the NBS simulates a landscape element, these often need additional dimension data. This data is stored in: './sources/setup/tables/nbs_properties.csv'. Check this table and adjust the values that correspond to the NBS you want to change.

WARNING for teracces/lynchets (graften), the vertical spacing in the table, should correspond to the vertical spacing in the applied map!

4) Run the data preparation again with you prefered workflow route – the new schematisation of the NBS will now be incorporated in the LISEM simulations.

Note: The schematisation does not have to cover the full Geul catchment, a schematisation for a subcatchment will work as well, just make sure that the starting map has the full Geul catchment extent and fill the rest of the map with 0 values.

Besides the spatial schematisation, also the parameter values or name/description of the NBS in the simulations can be changed. This can be done by adapting the values for the corresponding number in the table: './spatial_data/sources/setup/tables/lu_NBS_tbl.csv'

When after this change a new OpenLISEM simulation is made, these new settings will be applied to the input database.

3.1.2 Adding a new NBS that only effects landuse

If a NBS only changes landuse, new types can be added to the workflow.

1) Find the parameters with which you want to simulate this NBS for the parameters requested in ./sources/setup/tables/lu_NBS_tbl.csv

   a) Defining parameters will influence how the simulated NBS will perform in OpenLISEM, make sure you understand what the parameters mean, how OpenLISEM deals with them and how they fit within this calibrated model schematisation, to be able to make meaningfull simulations.

2) Make a map this the spatial schematisation of the NBS as described in 3.1.1, and place it in ./spatial_data/NBS_maps

3) Add a new row to the lu_NBS_tbl.csv. Under number fill a value > 10 and < 100, which is not yet used in the table for a different NBS.

4) Run the data_preparation again, and use your new NBS number to make the OpenLISEM simulations.

3.1.3 Adding a new NBS which also affects the DEM or other aspects of the landscape.

NBS that affect the DEM need specific code that implements their processes in the model. For example for swales, ditches with the correct volume are made in the DEM. When aiming to add such a measure, beside the steps in 3.1.2 also this specific process steps need to be included in the workflow. How to exactly do this depends on the desired NBS in combination with the possibilities of the OpenLISEM model.

For inspiration you can look around line 400 and down in ./sources/r_scripts/create_lisem_run.R; here the four currently implemented measures are coded. For each of these measures also an additional PCRaster script is used which is called from the R code.

3.2 Adding NBS scenarios

Adding an NBS scenario is much like adding a single NBS, with a few differences.

Note: In a scenario only measures can be used that are available as single NBS in the ./sources/setup/tables/lu_NBS_tbl.csv. All other NBS will automatically be set back to original landuse – without warning in the workflow!

1) Make a map with 0 = no NBS, nbs_number for all locations where the different NBS must be placed.

   a) Two options for lu_classes choices in the TIF map (figure 1):

      1) 'wrl' – numbering as used within the WRL project. Based on the table ./sources/setup/tables/transform_wrl_lu_classes.csv these are linked to the number as used in lu_NBS_tbl.csv

      2) 'def' – numbering according to lu_NBS_tbl.csv

b) Terraces need an additional map...

2) Place the map in ./spatial_data/NBS_maps/scenarios/ and let is start with number > 100 which is not already present in the folder. The name after the number will be used for the folder naming in the OpenLISEM simulations.

3) Run the data_preparation in the workflow again and prepare the scenario with the corresponding NBS number (> 100).

4. Advanced usage

4.1 Where to search for setup choices

In the setup of the OpenLISEM simulations many choices are made. We tried to place most of the choices in input tables to keep track of the all these choices. However some of the settings are also made within the code.

All files that relate to settings in the simulations are stored in the folder ./sources/ .

Some important tables are:

- selected_events.csv: this table contains the dates and begin and end times of rainfall events for the calibration. These can be changed to add other events, which will pe prepared when the workflow is executed for the option 'cal' = calibration. The precipitation and (optional) ndvi data has to be manually prepared.
- base_flow_cal_events.csv: for the calibration events a know baseflow can be added to the channel outflow points. If the baseflow is unknown automatically a no flow condition will be assumed
- transformations_maps.csv: the spatial_data folder contains vector and raster data input maps for the Geul catchment. These are transformed to PCRaster input maps for OpenLISEM. In this file several choices about the transformation are defined. I more input maps are needed, they can be added here.
- setup/outpoints_description.csv: this table contains all the coordinates of different 'outpoints' in the Geul catchment. Outpoints can be used to create a subcatchment within OpenLISEM. The exact coordinates have to be manually selected, see section 4.2.
- setup/runfile_template.run: this is a template runfile, which is loaded in R and where many options are updated for the different simulations.

However, some settings that are default for the whole simulation are manually set here.

In the folder ./sources/setup/calibration different .csv tables are available where multiplication factors for different parameters are defined to calibrate OpenLISEM for the whole Geul catchment. The folder ./sources/setup/tables contains .csv tables where landuse and NBS specific parameter choices are made.

4.2 Adding a catchment point for the interactive or manual workflow

Although the OpenLISEM dataset is prepared for the whole Geul catchment, it is often usefull to prepare a datset and simulation for a subcatchment. This reduces model runtimes and helps to make realistic detailed model inputs, which is harder for the whole Geul catchment.

In the workflow we have two appraoches to make subcatchment simulation datasets: a manual mode, which is used in the interactive or manual workflow, and a separate subcatchment devision for the hpc workflow. The two approaches differ in a few ways.

The manual subcatchments are defined in the file ./sources/setup/outpoints_description.csv. These points are placed within a channel/streamline and the upstream catchment area for this point is found using the local drain direction (ldd) map. For the HPC approach the subcatchments are predefined in the polygon map ./spatial_data/hpc_maps.gpkg; layer = lisem_subcatch. The main difference between these two appraoches is that the main channels/rivers are removed from the HPC subcatchments. This is done with the idea that OpenLISEM simulates the hillslope processes and then will be coupled to a river hydraulics model. Another difference is that the manual subcatchments have 1 outpflow point, which is often the default apporach in runoff analysis. The catchments made for the HPC have outflow over the whole length bordering the streams, this should be considered when analysing the results of the model runs.

To add a new catchment for the interactive or manual workflow, flow these steps:

1. Make sure the data_preparation for your desired resolution is executed. This should make a folder ./LISEM_data/Geul_[xx]m/maps. Which contains maps for the whole Geul catchment.
    a. With PCRaster and Nutshell you can execute pcraster commands to make new maps.

b. The command:
```
pcrcalc streams.map = streamorder(ldd.map)
```
executed in the prepared data folder, will make a raster map with the Strahler order of all streamlines.

2. When you view the map in for example QGIS, project it on a general topography map. Find you catchment of interest, and locate a cell with the highest strahler order at the outlet of the catchment.

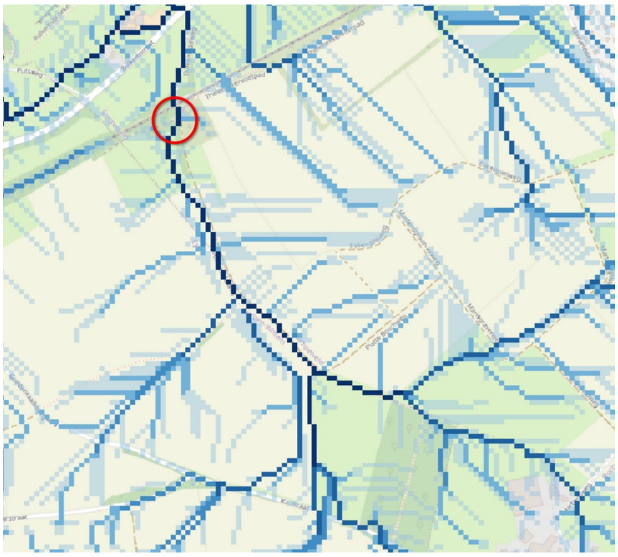

Figure 7: Example of Strahlerorder stream.map projected on topography. A potential good outlet point could be in the red circle.

3. Copy the x and y coordinates in RD NEW (EPSG:28992) format for this cell and add these in a new row in the table: ./sources/setup/outpoints_description.csv.

Note: In QGIS you can copy coordinates by 'right-clicking' a location on the map.

4. Fill all other columns, the catchment number should be a unique value, but the same for different resolutions (if applicable).

    a. The exact coordinates can vary per resolution, make the streams.map for each resolution of interest and find the correct coordinates. Assuming the same coordinates might work, but can also result in very small mini catchments if it doesn't allign well with the streams.map.

5. Rerun the data_preparation, after that you can use the new number of the subcatchment to prepare a OpenLISEM simulation for you catchment of interest.

4.3 Changing precipitation input files

In the current setup two sources of precipitation are used: observed events and standard events. The observed events are used for the calibration of the model, the standard events are used to explore the effects of NBS under varying conditions and are based on recurrence time analysis.

In the R script ./sources/r_scripts/prepare_rainfall_discharge.R all precipitation data is handeld. For observed rainfall, the radar data can be downloaded from the KNMI data portal (./sources/r_scripts/KNMI_precipitation.R) and transformed to OpenLISEM input. This combines a table with many columns corresponding to the rainfall radar 1km^2 cells, which are identified by the ID.map in the OpenLISEM datasets.

When using standard events, these are based on the table ./sources/setup/Buien_LISEM.csv. The code in section 1.5 of the rainfall preparation script will make functional OpenLISEM input files for all events in each colum of the file. In the current simulation setup only the events in colums `I` to `P` are used.

By adding a colum to the .csv file, addintional rainfall input files can be generated. To automatically include these in the workflow, check the code around line 690 in the script ./sources/r_scripts/create_lisem_run.R

4.4 Changing/updating spatial data input files

Maps in the folder spatial_data are used a basic input for the setup of the Geul catchment dataset. When switching to another region, these maps could all be replaced with data from a different catchment, most parts of the workflow will work directly, but some input tables might need adjustments.

When a map for the Geul catchment needs to be changed or updated, replacing the original map with a new map, and rerunning the data_preparation will include the changes in the model setup. Check the properties of how each map is transformed in the table ./sources/transformations_maps.csv.